



Hybrid composite *meta*-porous structure for improving and broadening sound absorption

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ABSTRACT

In order to improve and broaden the sound absorption performance of porous materials with low flow resistance, this study proposes the composite *meta*-porous structure embedded multiple lateral plates of different lengths, and investigates the corresponding sound absorption coefficients with the fixed and periodic boundary. Characteristics of spatial fluctuation and slow wave effect are embodied in acoustic pressure distributions. Further, the hybrid composite *meta*-porous structure with periodic boundary could achieve high sound absorption in range of 0–6.4 kHz. The Theoretical model using transfer matrix method suitable for these models is proposed, and verify the FEM results. Normalized surface impedance, phase and trajectory of the complex reflective coefficient illustrates the loss of acoustic energy inside the structure and leakage to the outside. By detailed discussion on the relationship between the geometric parameters and the corresponding sound absorption coefficients, the three boundaries would have different effects on sound absorption performance. Sound absorption test results by acoustic impedance tube confirm the rationality of the composite *meta*-porous structure with fixed boundary in range of 0.2–5.716 kHz and its practical application in the noise reduction.

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1. Introduction

The control of harmful noise and vibration has been a research focus for a long time [1–4]. Porous materials [5] have been playing a significant role in structural design, thermal shock, and sound absorption, such as the Noise, Vibration, and Harshness (NVH) acoustic packaging design for automobiles, acoustic materials for buildings, laminated panels for aerospace structures, gas explosion prevention, and armor protection. Conventional porous materials [6–10] have good sound absorption performance and low density, but the sound absorption effect is weak in the middle, even low, and sub-wavelength frequencies.

Different with conventional resonance structures [11–15], acoustic metamaterials [16–18] are a brand-new method that regulates sound propagation through periodic microstructures, among which the reflective [19] and absorptive [17,18,20] membrane metamaterials can achieve perfect sub-wavelength acoustic absorption. Concerning the perfect absorption of electromagnetic waves theory [21], the sub-wavelength perfect sound absorber based on the rigid material also gradually

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became a research hotspot [22–31]. Many composite and perfect hybrid absorbers, including labyrinth-type metastructure [22–28], Helmholtz resonator [29,30], and Fabry-Perot channels [31] show the novel physical properties of sub-wavelength absorption. However, it is difficult to achieve both sub-wavelength and broadband for the perfect rigid absorber with limited thickness. In the practical application of noise reduction, it is more meaningful to broaden the frequency range of high absorption (≥ 0.9), therefore, composite structures consisting of porous materials and rigid structures are effective methods to achieve broadband sound absorption [32–41]. Biomaterials, such as panels of porous materials, can improve the low-frequency sound absorption coefficient, but with limited efficacy at low frequencies only [33]. Periodically-arranged resonant inclusions [34] can improve the low-frequency sound absorption coefficient of porous-substrate materials. Micro-layer foam/film structure [35] increases the sound absorption coefficient of the single porous material by 2–3 times in a specific frequency range. Moreover, porous materials with inserted sheet [36] can improve the low-frequency sound absorption. Slit medium [37] combined with the porous material can broaden the sound absorption bandwidth below 500 Hz and achieve the 1/10 times sound wavelength of absorption effect at 300 Hz, and also form one or two new sound absorption peaks below 1.5 kHz by attaching to the film surface [38]. Combined polyurethane foam with barium titanate/nitrile butadiene rubber [39] can achieve the sound absorption bandwidth below 2 kHz. Also, both the superposition combination of phase gradient [40] porous material and multi-layer porous material [41] can improve the sound absorption. The 100 mm thick multi-layer porous material [41] can achieve broadband sound absorption within 8 kHz through the perfect sound absorption design, with the first perfect sound absorption point at 795 Hz; by reducing to 50 mm the thickness of the composite structure, the point moves at 1590 Hz.

This study proposes a novel composite *meta*-porous structure with fixed or periodic boundaries to overcome the difficulty of sub-wavelength sound absorption and achieve the target for the requirements of broadband sound absorption, which is different from previous rigid acoustic absorber [22–31] and membrane metamaterial [17–20]. The biggest advantage is that composite *meta*-porous structure could achieve the sub-wavelength and broadband sound absorption only by using lateral plates of different lengths, thus avoiding complicated spatial design and back cavities. This study enhances the research on broadband sound absorption of composite structures, which have potential applications in automotive NVH, sound absorption, and structural design of buildings.

The organization of this study is as follow. Section 2 proposes the design of composite *meta*-porous structure and verifies its good sound absorption performance. Section 3 explain the slow wave effect, present the hybrid composite *meta*-porous structure, discusses the impedance normalized surface impedance and reflection coefficient on complex plane. In the different models, the acoustic pressure distributions have the characteristics of spatial distribution. Section 4 sums up the effect of different geometric parameters on the sound absorption coefficients, and gives out the optimized structural size. Sample preparation and its determination of acoustic parameters are described in section 5. Section 6 displays the sound absorption test of composite *meta*-porous structure with the fixed boundary. Section 7 outlines the conclusion of this study together with the further works.

2. Composite *meta*-porous structure design

This section deals with the design of the composite *meta*-porous structure. Fig. 1(a) shows the periodic space arrangement of the proposed structure, where the gray part is the porous sponge, and the blue and red plates of different lengths run through each other in it.

In order to highlight the advantage of the proposed structure, this study compares the *meta*-porous structure with two models, S1 and S2 (Fig. 1(c) and (d)), respectively, both having one-side lateral plates, and a pure-porous model (Fig. 1(e)). The thickness of each lateral plate is $l = 0.001$ m, and the difference in Y-direction is $d = 0.001$ m. The left and right lateral plates (Fig. 1(f)) gradually extend from top to bottom along the X-direction according to the following pattern: $a_1 = b_1 = 0.008$ m, $a_i = a_{i-1} + d$ ($2 \leq i \leq 9$), $b_i = b_{i-1} + d$ ($2 \leq i \leq 9$); the left lateral plates are higher than the right ones in space. The distances between the left and right lateral plates and the porous sponge surface are $m = 4.9$ mm and $n = 7.4$ mm, respectively. The distance m between each lateral plate is fixed ($m = 5$ mm), the size of the whole unit cell is 30×50 mm ($W \times H$) on XY plane. According to previous studies [5,42], when fixed or periodic boundaries surround the composite *meta*-porous structure, the sound absorption is different. Therefore, this study adopts fixed and periodic boundaries to be attached to the composite *meta*-porous structure to describe the sound absorption characteristics of composite *meta*-porous structure comprehensively (see Fig. 1(g) and Fig. 1(h)). The fixed boundary condition corresponds to the normal incidence in the acoustic impedance tube, while the periodic boundary condition corresponds to the incidence at any angle in the free field. Note that no matter what kind of boundary the models have, there are rigid backings behind, which is also consistent with most cases in engineering applications [5,41].

3. Sound absorption characteristics

3.1. FEM calculation

FEM is an effective numerical method [11–14,37] of acoustic prediction and calculation for complex internal geometric structures, whose accuracy has been confirmed by theoretical solutions and experiments. This study calculates the

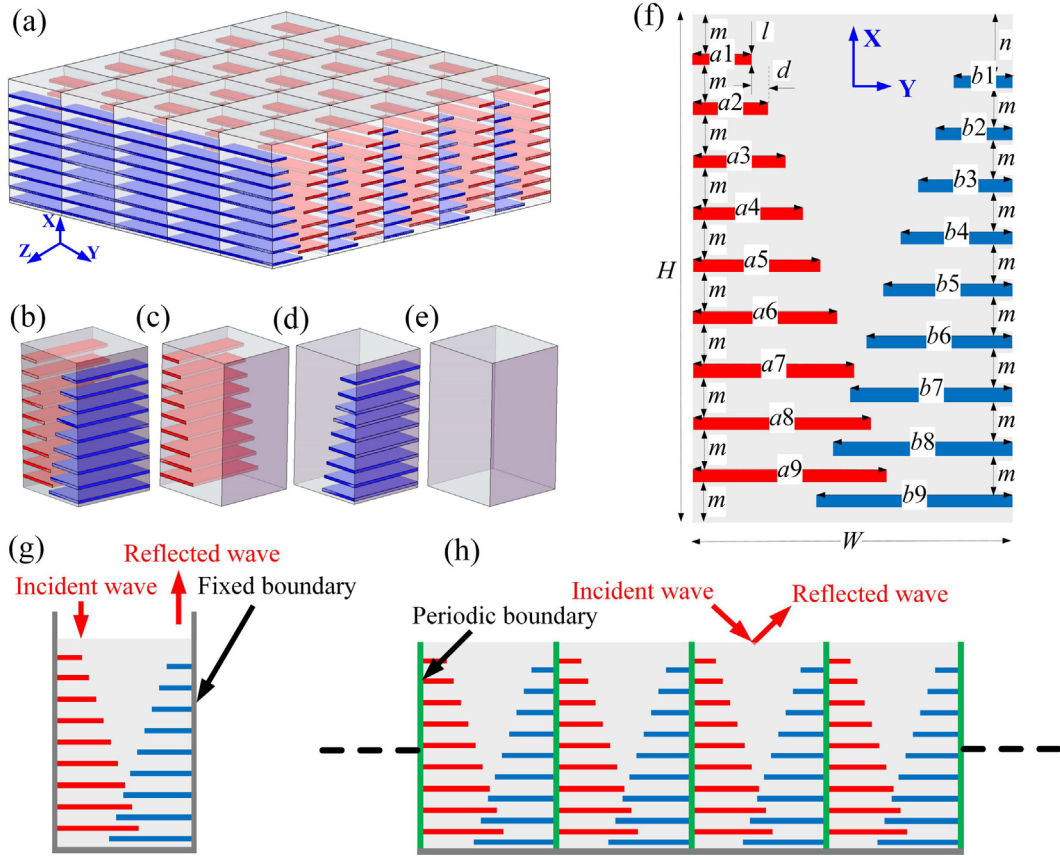


Fig. 1. Schematic of the composite *meta*-porous structure. Spatial periodic arrangement (a) and unit cells: composite *meta*-porous (b), comparison model S1 (c), comparison model S2 (d), pure-porous model (e); (f) Cross-section of the proposed structure on the Y-X plane; (g) Fixed boundary arrangement; (h) Periodic boundary arrangement.

composite *meta*-porous structure through the thermoviscous module of COMSOL Multiphysics® Modeling software [44]. The linearized Navier-Stokes equation, the continuity equation, and the energy equation are solved in the thermoacoustic domain, and the Helmholtz equation is solved in the acoustic domain. Floquet periodic boundary conditions is used for periodic boundary, and sound hard boundary is for fixed boundary which represents that the acceleration (and thus the velocity) is zero. The perfectly matched layer (PML) provides absorption of propagating waves without introducing reflections from the interface between the PML and the physical domain. The FEM simulation adopts the custom mesh, with the minimum mesh cell less than $1/6$ of the corresponding wavelength. The upper boundary is the incident plane wave, and the incident acoustic pressure magnitude is 1 Pa.

In recent years, the semi-empirical five-parameter Johnson Champoux Allard (JCA) model has been widely used, aiming at describing the acoustic propagation, acoustic performance, and structural optimization in porous materials [5]. When the incident acoustic pressure level is not high, the porous sponge is feasible to adopt the JCA model [5]. Here, the five acoustical parameters used are porosity $\varphi = 0.96$, tortuosity $\alpha_{\infty} \approx 1.07$; viscous and thermal characteristic lengths Λ and Λ' are $273 \mu\text{m}$ and $672 \mu\text{m}$, respectively, and flow resistance $\sigma = 2843 \text{ Pa}\cdot\text{s}/\text{m}^2$ [45,46]. Note that the flow resistance φ of the porous sponge in this study is lower than traditional sound-absorbing materials $10^3 \leq \sigma \leq 10^4 \text{ Pa}\cdot\text{s}/\text{m}^2$ [46]. In FEM calculation, the porous media acoustic section of thermoviscous module is used to simulate the porous sponge, and noting that the porous matrix is approximately rigid when the incident sound pressure is not very high.

3.2. Sound absorption of composite *meta*-porous structures

Fig. 2 shows the FEM calculation results of sound absorption coefficients of composite *meta*-porous structure, comparison model S1, comparison model S2 and pure-porous model at different boundaries. Table 1 summarizes the coordinates of points O1-O8 in Fig. 2 (a) and (b).

The composite *meta*-porous structure has good sound absorption characteristics in the range of 0–6.4 kHz, whether the fixed or periodic boundary encloses it. The overall sound absorption performance of the proposed structure is significantly better than that of the two comparison models S1 and S2, and the pure-porous model. The composite *meta*-porous structure

with fixed boundaries (Fig. 2(a)) only generates three sound absorption peaks below 3 kHz, but with sound absorption coefficients all higher than 0.9, while both the comparison models and pure-porous model have only one single lower sound absorption peak in the same frequency range. In the range of 3–6.4 kHz, the composite *meta*-porous structure has multiple sound absorption peaks and distinct valleys, and the sound absorption coefficients are slightly weaker than S1 and S2 models. The first sound absorption peak O1 is at 805 Hz, corresponding to the sound absorption coefficient up to 0.9, much better than comparison model S1 (0.45), S2 (0.4), and pure porous model (0.32), which leads to the sub-wavelength sound absorption effect of 1/8.5 times. The composite *meta*-porous structure with periodic boundary has four sound absorption peaks in the range of 0–6.4 kHz, more than three of the two comparison models, and the two peaks of the pure porous model. Its first absorption peak O5 is at 1065 Hz and also has the sub-wavelength sound absorption effect of 1/6.4 times. In Fig. 2 (a), the average sound absorption coefficient of composite *meta*-porous structure is up to 0.8126 in range of 0–6.4 kHz, while that of comparison model S1 and S2 are only 0.7536 and 0.764, respectively. The worst average absorption coefficient 0.5961 belongs to pure-porous model. Further observation of Fig. 2 (b) shows that the composite *meta*-porous structure has the best broadband sound absorption effect whether it has the fixed convenient or periodic boundary. Fig. 2 (a) and (b) highlight the small effect of two different boundaries on the sound absorption performance, which enlarges in the case of an internal and complicated structure. The composite *meta*-porous structure moves the first sound absorption peak to the low frequency and

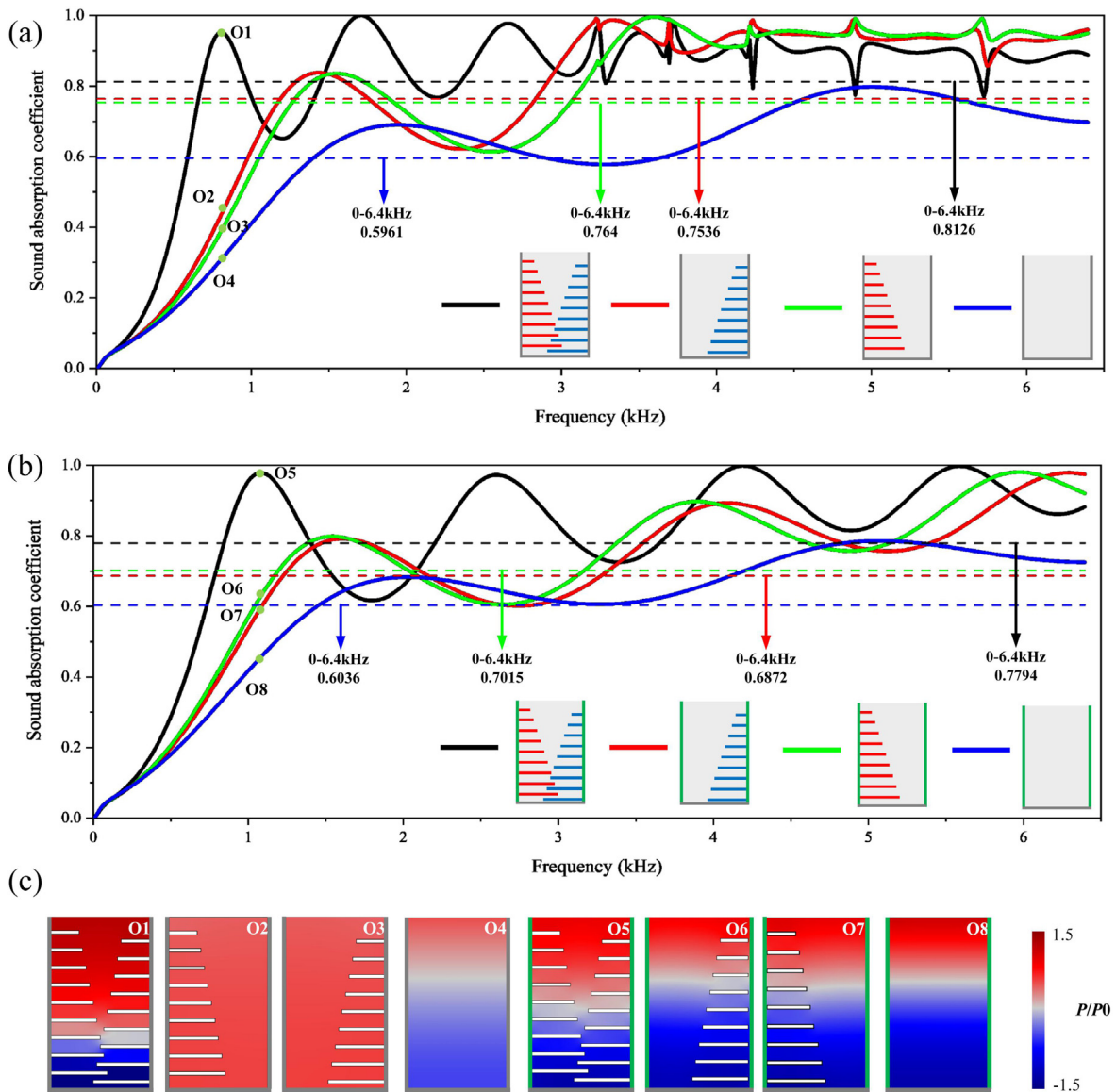


Fig. 2. The sound absorption coefficient of the composite *meta*-porous structure compared to S1 and S2 models, and pure porous model with fixed (a) and periodic (b) boundary. (c) Acoustic pressure fields P/P_0 distributions of points O1–O8. P_0 is the pressure of the incident sound wave.

Table 1

Coordinates of points in Fig. 2 (a) and (b).

Points	O1	O2	O3	O4	O5	O6	O7	O8
Frequency (Hz)	805	805	805	805	1065	1065	1065	1065
Sound absorption coefficient	0.95	0.45	0.4	0.32	0.98	0.6	0.55	0.42

generates more peaks at medium and high frequencies. Comparing the composite structure with S1 and S2 models, it can be seen that structural overlap at the bottom will lead to induce coupling modes for resonant absorption peaks, and this peak would be raised and moved to the low frequency. Fig. 2 (c) shows the distribution of the acoustic pressure field at points O1–O8. At the same frequency of 805 Hz, the composite *meta*-porous structure can effectively lose sound energy, and with the increase of depth, the acoustic pressure gradient changes and gradually weakens. The acoustic pressure distribution in the two comparison models shows no such spatial fluctuation.

Given that at sub-wavelength frequency [42], only part of the sound waves could enter the pure porosity and be absorbed, the sound absorption effect of the pure porous model is limited. By observing the acoustic pressure distribution at points O5–O8, the sound absorption region in the composite *meta*-porous structure is more extensive, and it has the high sound absorption coefficient 0.98. Points O5–O8 are located at 1065 Hz; their sound absorption coefficients are still higher than points O1–O4 located at 805 Hz. Under the periodic boundary condition, the propagating waves in the composite *meta*-porous structure can enter the adjacent unit cells without the obstruction of the fixed boundary, which means the acoustic pressure and particle velocity are continuous at the connection surface between each other. Therefore, the coupled propagating waves bring higher or lower the sound absorption performance, but is also depends on the internal structure. Compared with the fixed boundary, the composite *meta*-porous structure with periodic boundary reaches its first sound absorption peak more slowly in the frequency domain, with higher peak-to-peak values.

No matter what kind of boundary conditions attached to composite *meta*-porous material, the acoustic pressure distributions of the first sound absorption peaks also have spatial fluctuation characteristics, which must be related to the spatial distribution of the lateral plates. Therefore, this study chooses the end part (points 1–4) of the fourth and fifth layers of the left and right lateral plates distributed near the center of the unit cells as the object of research (Fig. 3(a) and Fig. 3 (e)). Below 0.5 kHz, the four points in the composite *meta*-porous structure with fixed boundary all have high acoustic pressure, but the corresponding particle velocities are small. In the range of 0.5–2.5 kHz, the acoustic pressure of points 1 and 2 is more significant than that of points 3 and 4, but the particle velocity is smaller, indicating that the normalized equivalent acoustic speed drop is lower, corresponding to the two valley values in Fig. 3 (d). Note that this process does not consider the effect of the near-field evanescent wave. Similarly, in 4.5–6.4 kHz range, the acoustic pressure of points 3 and 4 is more significant, but point 2 has smaller particle velocity, therefore, the two valley values in Fig. 3 (d) are also evident. In the range of 0–6.4 kHz, the equivalent acoustic velocity magnitudes in Fig. 3 (d) are mostly lower than the sound velocity in air, showing that the distribution of lateral plate could bring the slow sound effect [47], which enlarges the contribution of sound energy loss. In Fig. 3 (f) and (g), the acoustic pressure of points 1 and 2 is still larger than points 3 and 4 in the range of 1–2.5 kHz, but at the higher frequency, the acoustic pressure of points 1–4 varies with each other. The particle velocity of points 1 and 3 is high, so only points 1 and 3 have the slow sound effect. The equivalent acoustic velocity of the four points in Fig. 3 further shows that the spatial fluctuation characteristics of acoustic pressure in the composite *meta*-porous structure are different with different boundary conditions, which depends on the adjustment of the lateral plate and the combination of boundary conditions.

3.3. Sound absorption of hybrid composite *meta*-porous structures

In this section, the sound absorption coefficients of the composite *meta*-porous structure at fixed and periodic boundaries are compared, and hybrid composite *meta*-porous structure is proposed.

Fig. 4 (a) compares the sound absorption curves of the composite *meta*-porous structure for different boundaries, the average sound absorption coefficient of fixed boundary has more advantages. From black and red lines in Fig. 4(a) and Table 2, an interesting phenomenon is that some sound absorption peaks (Points L1 and L6) under fixed boundary conditions approximately correspond to the sound absorption valleys (Points L2 and L7) under periodic boundary conditions in the frequency domain. Next, this study tries to build the hybrid composite *meta*-porous structure with two different boundaries. Within a large unit cell, the fixed boundaries are inserted between each small unit cell, the left and right sides of the large unit cell are periodic boundaries. When the number of the fixed boundary in the large unit cell increases, the values of the original first and second sound absorption valley increase (Points L3, L4 and L5 in Fig. 4(b)). In the range 4–6.4 kHz, due to the fluctuation of the sound absorption curve decreases, the sound absorption coefficients are mainly between 0.9 and 1. From the perspective of practical application, the hybrid composite *meta*-porous structure has the stable acoustic absorption effect at high frequencies because of the little fluctuation and maintains the good sub-wavelength acoustic absorption effect at medium or low frequencies. To simplify the representation, from now on, one refers to the hybrid composite *meta*-porous structures as the H1, H2, and H3 models. From average sound absorption coefficient, H3 model is better than H2 model, and H2 model is better than H1 model, but the difference between them is not large in range of 0–6.4 kHz. The black, red

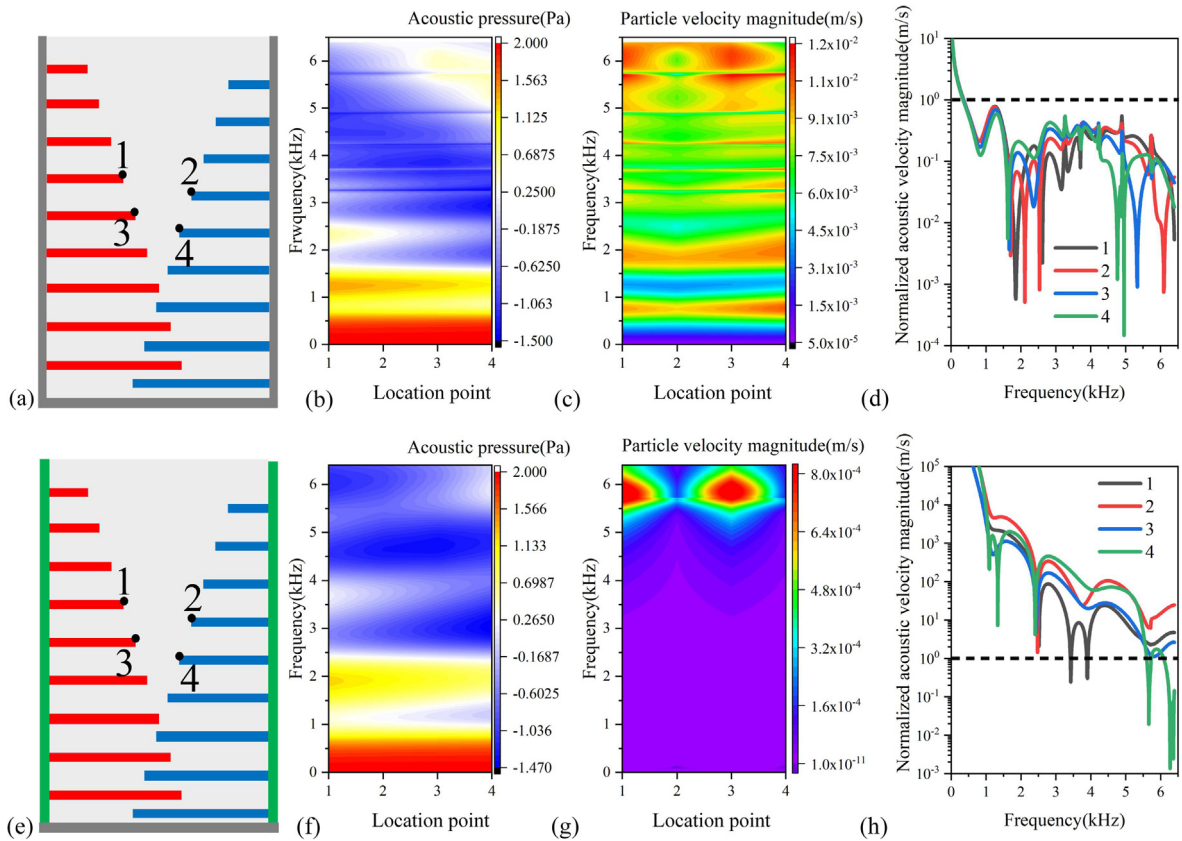


Fig. 3. Location of Points 1–4 on the composite meta-porous structure, and the corresponding acoustic pressure, particle velocity magnitude, and normalized acoustic velocity magnitude: (a)–(d), fixed boundary; (e)–(h), periodic boundary.

and blue lines in Fig. 4(c)–(h) represent the corresponding models, H1–H3 models. The sound absorption curve can only express the absorption result of the model to the incident sound energy. Hence, in order to give the damping characteristics of the surface particle velocity, Fig. 4 (c) and (d) show the real part and the imaginary part of the normalized surface impedance of H1, H2, and H3 models, respectively. Under the condition of high sound absorption or even perfect sound absorption [28,29], the surface impedance of the models must match the background medium very well; that is, the real part of the normalized impedance is close to 1, while the imaginary part tends to zero. Impedance mismatching makes part of the acoustic energy unable to enter the porous sponge, thus weakening the sound absorption effect of the model. In Fig. 4 (b), the first two sound absorption peaks P1 and P2 of the H2 model are quasi-perfect absorption points, and the corresponding absorption coefficient reaches 0.99. Therefore, at the P1 and P2 points, the surface impedance of the H2 model is closer to that of the air. The sound absorption coefficient and the surface impedance are not enough to wholly evaluate the relationship between the inner losses and the leakage of the acoustic energy in the three models. Therefore, Fig. 4 (e) shows the phase relationships of the reflection coefficients on the complex plane. At P1 and P2, the phase change of the reflection coefficient of the H2 model is opposite and close to π , so this is the critical condition to achieve a high absorption point [48]. Furthermore, from Fig. 4 (e), below 3 kHz, the H1 model has more considerable leakage than inner losses, and it belongs to the case of lack, while the leakage of H3 is smaller, belonging to the case of excess. Since the significant phase fluctuation of the reflection coefficient of the H3 model, more absorption peaks can be generated [48]. Fig. 4(f)–(g) show the Argand diagram of the reflection coefficient on the complex plane, and corresponding frequency ranges are 0–6.4, 0–1.5, and 1.5–3 kHz, respectively.

The phase analysis of wave scattering widely uses the Argand diagram to determine the existence of the resonance state, and also it can determine the perfect resonance absorption of sound energy in the system [50]. In the Argand diagram, when the loss of the model changes, the trajectory of the reflection coefficient changes to an irregular circle around the absorption peak frequency. The diameter of the circle is inversely proportional to the loss. For the H1, H2, and H3 models, the lateral plates located in the porous sponge induce some isolated resonance points below 6.4 kHz corresponding to the isolated small rings in Fig. 4(f)–(h). Three different color arrows in Figs. (f)–(h) delineate the reflective coefficient trajectory from low to high frequencies. When the losses are too high, the reflection coefficient trajectory encloses an area that reduces. Therefore, the trajectory area of the reflection coefficient of the H1 model is large, while that of the H3 model is the smallest, which belongs to the case of excess. Below 1.5 kHz, the reflection coefficient trajectory of the H2 model is very close to the point (0,0), indi-

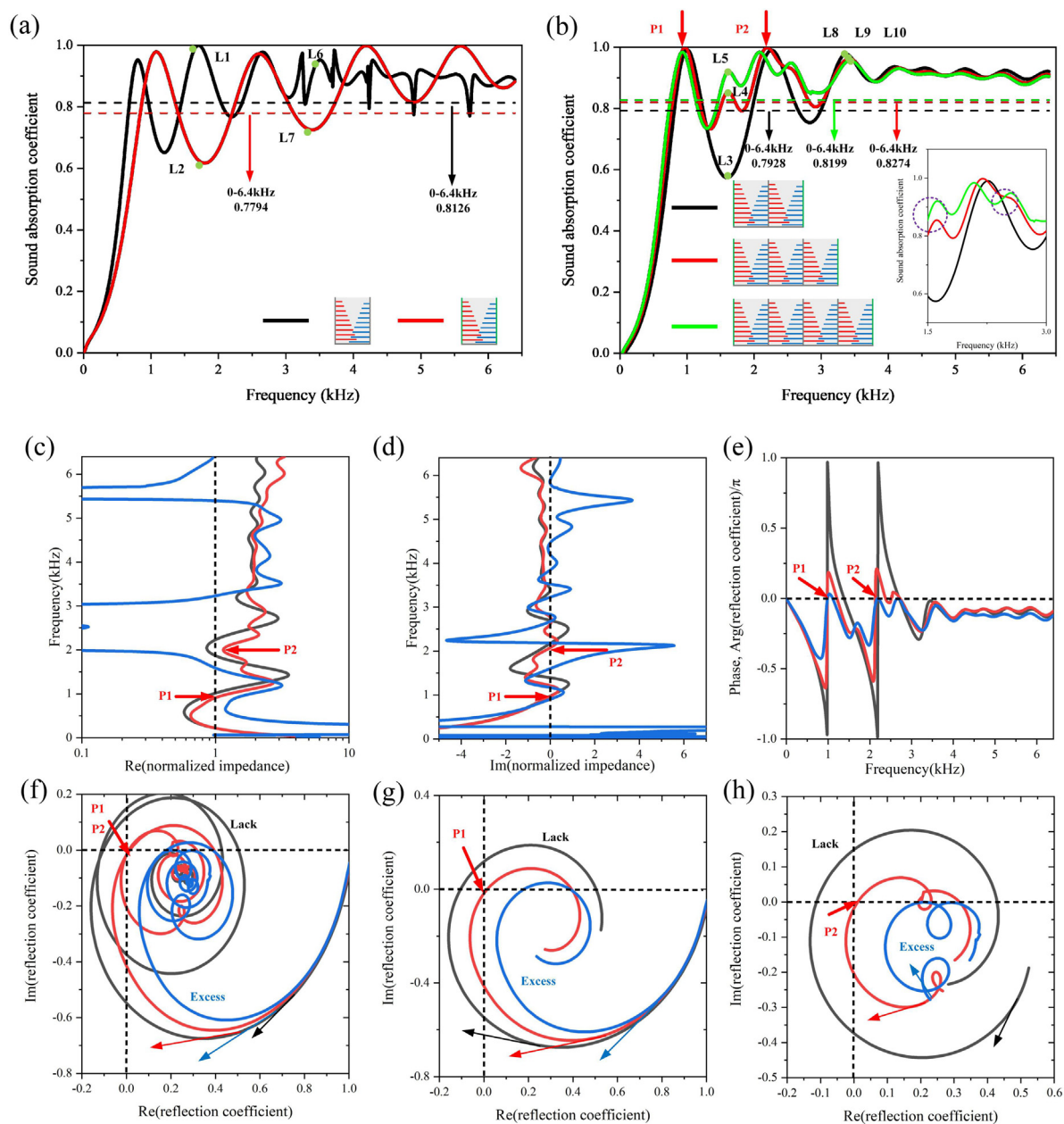


Fig. 4. The sound absorption coefficients, normalized surface impedance, and reflection coefficient on the complex plane. Sound absorption coefficients (a) of composite *meta*-porous structure between fixed and periodic boundary. Sound absorption coefficients (b) of the three different hybrid composite *meta*-porous structure. Normalized surface impedance: (c) real part, (d) imaginary part. Reflective coefficient phase change (e) and Argand diagram (f)–(h) of complex reflection coefficient trajectory.

Table 2

Coordinates of points in Fig. 4 (a) and (b).

Points	L1	L2	L3	L4	L5	L6
Frequency (Hz)	1692	1842	1579	1579	1579	3502
Sound absorption coefficient	0.91	0.62	0.58	0.85	0.92	0.95
Points	L7	L8	L9	L10	P1	P2
Frequency (Hz)	3406	3435	3421	3371	968	2191
Sound absorption coefficient	0.73	0.96	0.96	0.98	0.99	0.99

cating the quasi-perfect absorption case. At the same time, the reflection coefficient trajectories of H1, H2, and H3 models are not distorted, and every value corresponds to the separate absorption peak, which is consistent with the conclusion in Fig. 4 (e).

In the range of 1.5–3 kHz, the reflection coefficient trajectory of the H1 model is still not distorted, but the reflection coefficient trajectory of models H2 and H3 mostly produces two small isolated circles. Furthermore, the isolated small circle of H3 model has a large radius, which means that its coupling strength is high, so the corresponding two absorption peaks are also higher than that of the H2 model, as shown in the purple dotted circle in Fig. 4(b). When the model itself is to be changed, one should combine the complex plane analysis of the reflection coefficient of the structure surface, so that one can adjust the structural design comprehensively from the balance between the loss, the leakage, and the isolated resonance coupling strength, to achieve the better absorption target. Previous studies on different types of sound absorption acoustic metamaterials most used the method of adjusting the structural geometry [23,28,29,43,48–50] or adjusting porous materials to achieve the single- or multiple-frequency sound absorptions [41]. The critical coupling of the model system is also the key to achieve perfect sound absorption, but the effect of different boundaries has not been reported. Points P1 and P2 are close to the perfect sound absorption, which means that the coherent losses and leakage of the H2 model are close to equilibrium. The inherent losses of the H3 model are too high to achieve perfect sound absorption, but there are more phase changes of the reflection coefficient, which can also achieve more relatively high sound absorption characteristics, that is, the sound absorption coefficient is higher than 0.9. Therefore, for the infinite composite *meta*-porous structure in the periodic arrangement, adding the fixed boundary between each small unit cell is simple and quick, it can make the whole is in the case of lack or excess without totally updating the geometry or materials, which is a simple and quick method to adjust the loss.

Fig. 5 shows the acoustic pressure distribution P/P_0 at points L1–L10, refer to Fig. 4 (a) and (b) for the location of the points. Compared with points L2 and L7, the acoustic pressure of points L1 and L6 has noticeable spatial fluctuation. Further observation of points L3–L5 shows that the acoustic mode of point L1 is not embodied in point L3, but appears in points L4 and L5. The acoustic modes of points L3 and L2 are identical, so the corresponding sound absorption coefficients do not differ too much. The acoustic modes of point L6 are embodied in points L8–L10, which indirectly proves that the sound absorption coefficients of points L8–L10 will be increased. Therefore, it is necessary to combine more acoustic modes of high sound absorption points to improve the average sound absorption performance of the hybrid composite *meta*-porous structure. Under the condition of different boundaries and frequencies, the composite *meta*-porous structure itself has different absorption regions for sound energy, so when the thickness of the model is constant, the acoustic pressure can be spatially modulated by adjusting the lateral plates, and further affects the sound absorption characteristic. The combination of the porous material and the rigid structure is robust, simple, and flexible.

Whether for resonance sound absorber [11–15,22–29] or porous material [5,33–42,45,46], oblique incidence and normal incidence of sound waves have the great different effect on the sound absorption coefficient [5,37,42,49]. In Fig. 6(a), when the incidence angle increases from 0° to 30°, the sound absorption peaks below 3 kHz raise, but there are still some fluctuations at high frequency. When the incidence angle is 45° and 60°, high-frequency fluctuations disappear, and the first sound absorption valley rises. From Table 3, the composite *meta*-porous structure with fixed boundary achieves two quasi-perfect sound absorption points T1 and T2 at the incidence angle of 60°. The worst overall sound absorption performance occurs at the 75° incidence angle. This result shows that the continuous increase of incident angle does not always improve the sound absorption performance, and there should be an angle threshold for different structure models which are also reflected by average sound absorption coefficient in Fig. 6 (a). At the 60° incidence angle, the average sound absorption coefficient reaches 0.8993, which is much higher than 0.8126 at the 0° incidence angle. But the average sound absorption coefficient is only 0.8168 at the 75° incidence angle. For the composite *meta*-porous structure with periodic boundary, the first sound absorption peak moves to the higher frequency when the incidence angle increases, while the remaining peaks gradually disappear. At the 60° incidence angle, the number of absorption peaks are markedly reduced below 6.4 kHz. When the incidence angle is 75°, the sound absorption coefficient of point T3 at 721 Hz is 0.84. Different from Fig. 6 (a), the average sound absorption coefficient in Fig. 6 (b) increases with incident angle, this relationship is more distinct for low-mid frequencies

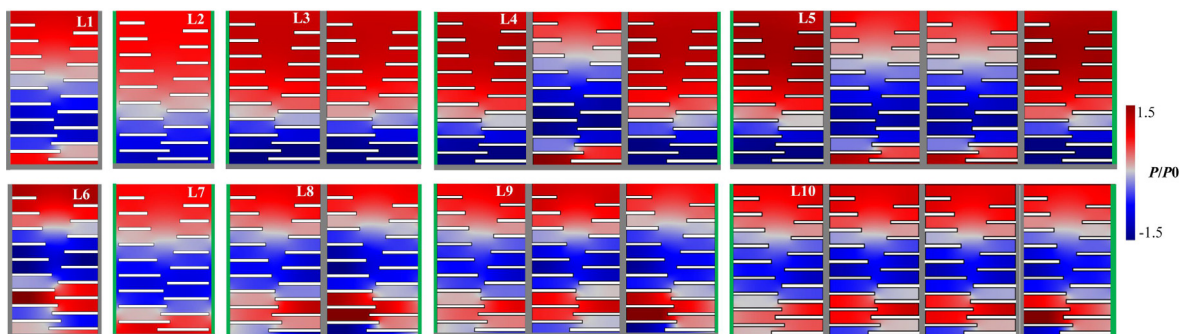


Fig. 5. Acoustic pressure fields P/P_0 distributions at points L1–L10.

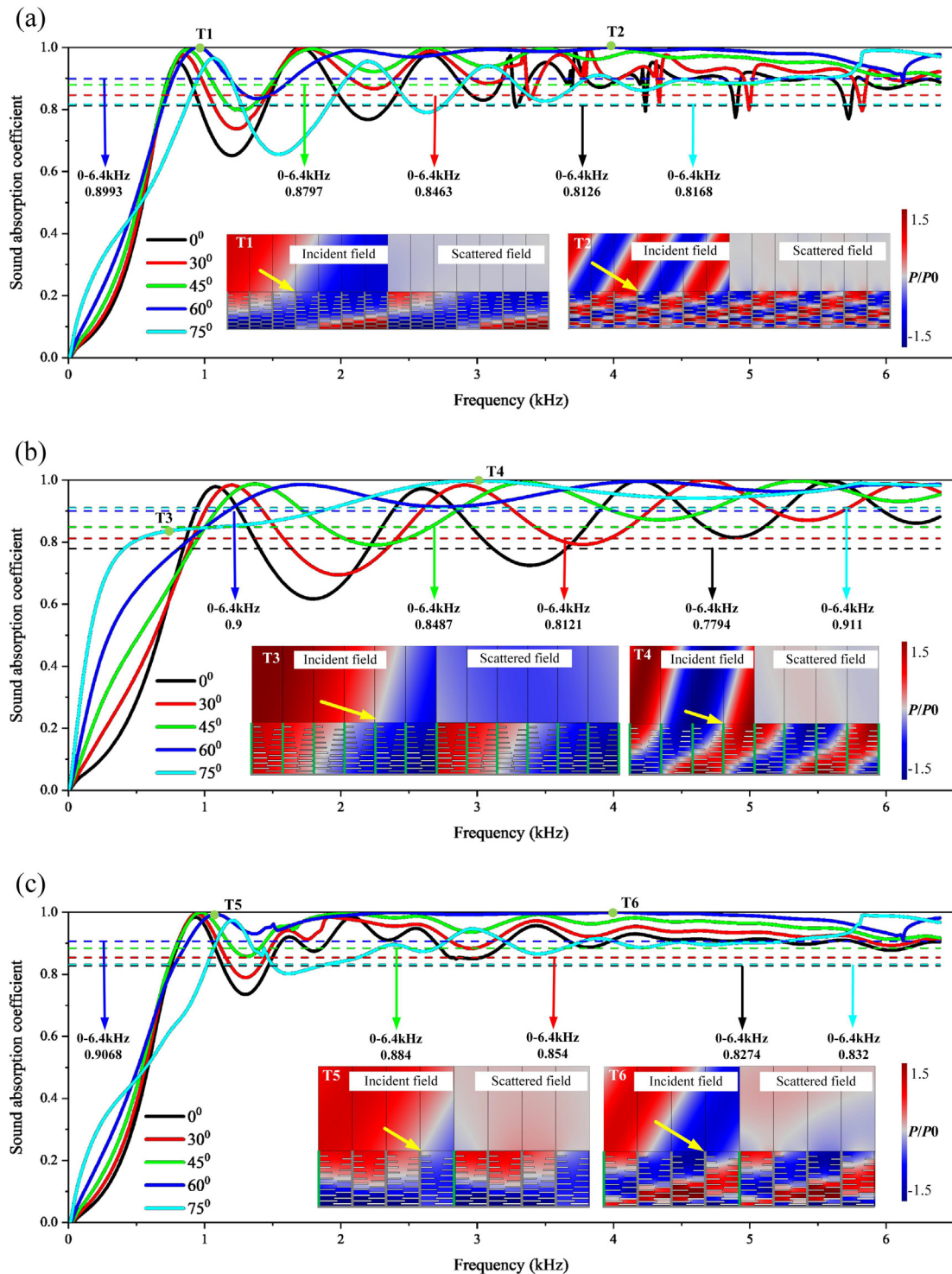


Fig. 6. Sound absorption coefficient and acoustic pressure field distribution varying the incidence angle in the range of 0° – 75° . (a) Composite *meta*-porous structure with fixed boundary; (b) Composite *meta*-porous structure with periodic boundary; (c) H3 model with periodic boundary. Yellow arrows delicate the incident wave direction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

than for high frequencies. For most equipment and automobile, the vibration noise in the range of 300–500 Hz is a generally challenging problem. The composite *meta*-porous structure with periodic boundary shows the significant advantage in the case of oblique incidence with large inclined angle, indicating that such this structure has selective filtering characteristics.

Table 3
Coordinates of points in Fig. 6.

Points	T1	T2	T3	T4	T5	T6
Frequency (Hz)	944	3983	721	2939	1080	4032
Sound absorption coefficient	0.99	0.99	0.84	0.99	0.99	0.99

For the H3 model, when the incidence angle increases from 0° to 60° , the first sound absorption peak moves weakly to the high frequency, and the average sound absorption coefficient rises from 0.83 to 0.91 in the range of 0–6.4 kHz. At the incidence angle of 75° , the whole sound absorption coefficient decrease, and the corresponding sound absorption coefficient is only 0.832. It should be noted that the increase of the average absorption coefficient is proportional to the incident angle within about 500 Hz. In addition, the increase of the incident angle will also make the first sound absorption peak move to high frequency. The acoustic pressure distribution in Fig. 6 suggests that the acoustic modes of two different models are different, which is also the reason for the different sound absorption characteristics. In the composite *meta*-porous structure with periodic boundary, the propagation wave is similar to the plane wave with distortion, because of the influence of the lateral plate as a rigid filler. However, from points T5 and T6, the fixed boundary additionally affect the propagation wave in the H3 model, where the acoustic modes are more abundant.

3.4. Transfer matrix method combined with effective medium model

In order to verify the correctness of the above FEM results and provide the theoretical model of the sound propagation inside the composite *meta*-porous structure, this study adopts the equivalent medium model [5,47] and the transfer matrix (TM) method [5,49].

From Fig. 7, the plane wave is incident to the surface of the composite *meta*-porous structure at an angle ϕ , and its surface impedance is z_{s1} . Set the wave number k_0 in the air to obtain:

$$k_0 = \sqrt{k_{0,x}^2 + k_{0,y}^2} \quad (1)$$

where $k_{0,x}$ and $k_{0,y}$ are wave number components at the X- and Y-directions, respectively.

The impedance in the air is z_0 , acoustic pressure $p_{0,b}$, and velocity $u_{0,b}$. According to the geometric features of the composite *meta*-porous structure, the composite *meta*-porous structure are divided by nineteen effective medium layers, as shown in Fig. 7. The first effective medium layer is porous sponge, where the acoustic pressure is $p_{1,t}$ and velocity $u_{1,t}$ at the interface where the surface impedance is z_{s1} . Between the first and second effective medium layers, the surface impedance is z_{s2} , the pressure is $p_{1,b}$ and velocity $u_{1,b}$. Also, wave number in porous sponge is k_1 , the impedance of the first layer of porous sponge is z_1 , and

$$k_1 = \sqrt{k_{1,x}^2 + k_{1,y}^2} \quad (2)$$

where $k_{1,x}$ and $k_{1,y}$ are the wave number components at the X- and Y-directions, respectively.

According to the acoustic Snell's law [5], there is the corresponding relationship between reflection and transmission

$$k_{0,y} = k_0 \cdot \sin\phi = k_1 \sin\epsilon_1 \quad (3)$$

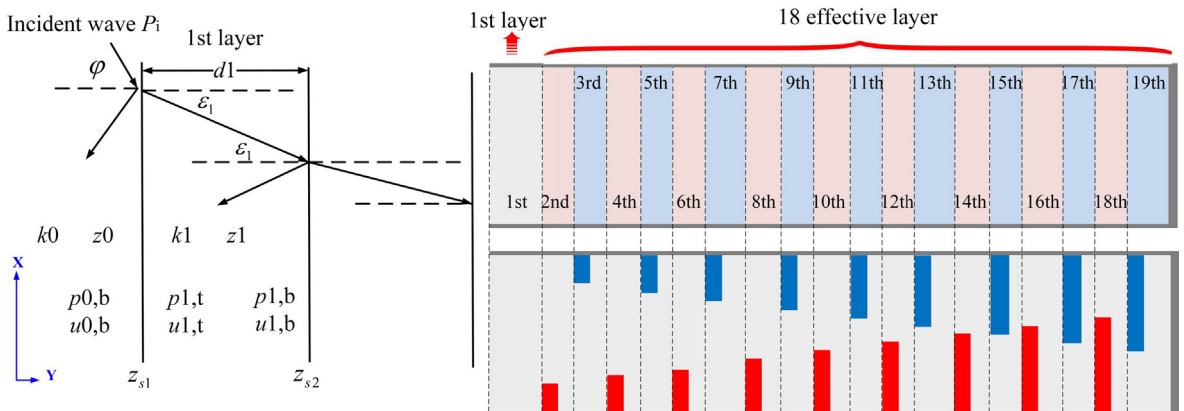


Fig. 7. Schematic of the transfer matrix (TM) method.

where ε_1 is the refraction angle. Due to the anisotropic spatial distribution, the acoustic properties of the porous sponge depend on fiber orientation. Here, the wave number perpendicular to the fiber is k_N , the wave number parallel to fiber is k_P , and the following relationship exists [43]:

$$k_{1,x}^2 = k_N^2 \left(1 - \frac{k_{1,y}^2}{k_P^2} \right) \quad (4)$$

Due to the following correspondence:

$$k_N = \omega \sqrt{\frac{\rho_{1,e}}{K_{1,e}}} \quad (5)$$

The JCA model gives the equivalent density $\rho_{1,e}$ and stiffness $K_{1,e}$ of the first layer of porous sponge. The JCA model regards the porous sponge and internal air as an equivalent fluid, and the sound absorption mechanism is mainly viscous effect and temperature effect, which correspond to two parameters, the effective density $\rho_{1,e}$ and bulk modulus $K_{1,e}$ can be obtained as:

$$\rho_{1,e} = \frac{\rho_0 \alpha_\infty}{\phi} \left(1 + i \frac{\omega_v}{\omega} F(\omega) \right) \quad (6)$$

$$K_{1,e}^{-1} = \frac{\gamma P_0}{\phi \left(\gamma - (\gamma - 1) (1 + i(\omega'_c / (\text{Pr}\omega)) G(\text{Pr}\omega))^{-1} \right)} \quad (7)$$

$$F(\omega) = \sqrt{1 - i\eta\rho_0\omega \left(\frac{2\alpha_\infty}{\sigma\phi\Lambda} \right)^2} \quad (8)$$

$$G(\text{Pr}\omega) = \sqrt{1 - i\eta\rho_0\text{Pr}\omega \left(\frac{2\alpha_\infty}{\sigma'\phi\Lambda'} \right)^2} \quad (9)$$

where air density $\rho_0 = 1.213 \text{ kg/m}^3$, Prandtl number $\text{Pr} = 0.71$, atmospheric pressure is P_0 , specific heat ratio $\gamma = 1.4$, dynamic air viscosity $\eta = 17.9 \times 10^{-6} \text{ Pa}\cdot\text{s}$. The Biot frequency is ω_v ; adiabatic cross-over angular frequency is ω'_c , and thermal resistivity σ' used in Eq. (6) and (7) are as follows:

$$\omega_v = \sigma\phi / \rho_0 \alpha_\infty \quad (10)$$

$$\omega'_c = \sigma'\phi / \rho_0 \alpha_\infty \quad (11)$$

$$\sigma' = 8\alpha_\infty \eta / \phi \Lambda' \quad (12)$$

This study adopts the relationship between flow resistances of parallel σ_N and vertical σ_P [50,51] to describe the different wave numbers of parallel k_P and vertical k_N in the porous sponge: $\sigma_N \approx A\sigma_P$, where A is 0.5 or 0.6. Assume that only plane waves exist in the model, acoustic pressure and the particle velocity are the continuous at both surfaces of the first layer of porous sponge. Hence, there are the following relationships in the first layer of porous sponge:

$$\begin{bmatrix} p_{1,t} \\ u_{1,t} \end{bmatrix} = \begin{bmatrix} \cos(k_{1,x} \cdot d_1) & j \frac{\omega \rho_{1,e}}{k_{1,x}} \sin(k_{1,x} \cdot d_1) \\ j \frac{k_{1,x}}{\omega \rho_{1,e}} \sin(k_{1,x} \cdot d_1) & \cos(k_{1,x} \cdot d_1) \end{bmatrix} \begin{bmatrix} p_{1,b} \\ u_{1,b} \end{bmatrix} \quad (13)$$

The surface impedance z_{s1} of the first layer of porous sponge is [42]:

$$z_{s1} = \frac{-jz_{s2}z_1 \frac{k_{1,x}}{k_{1,x}} \cot(k_{1,x} \cdot d_1) + \left(z_1 \cdot \frac{k_{1,x}}{k_{1,x}} \right)^2}{z_{s2} - jz_1 \frac{k_{1,x}}{k_{1,x}} \cot(k_{1,x} \cdot d_1)} \quad (14)$$

After dividing the composite *meta*-porous structure into 18 layers for calculation (Fig. 7), the surface impedance z_{s1} - z_{s19} of each layer can be calculated, so as long as the equivalent density $\rho_{1,e}$ - $\rho_{19,e}$ and stiffness $K_{1,e}$ - $K_{19,e}$ of each layer are obtained, and the surface impedance and sound absorption coefficient of composite *meta*-porous structure can be calculated [42].

The TM aims at the infinite structure in half-space [5,42], so to simulate the periodic boundary. However, for fixed boundary [47,49], this study establishes the corresponding equivalent medium model (Fig. 8). Layer i ($2 \leq i \leq 19$) is composed of medium A_i and medium B_i . Layer 1 and medium B_i are porous sponges, whose equivalent density and equivalent modulus can be calculated from Eq. (6)-(10). By considering medium A_2 , one obtain the equivalent density $\rho_{2,e,y}$, $\rho_{2,e,x}$ along the Y- and X-directions, and the equivalent bulk modulus $K_{2,e}$ [43] as follows:

$$\rho_{2,e,y} = \infty \quad (15)$$

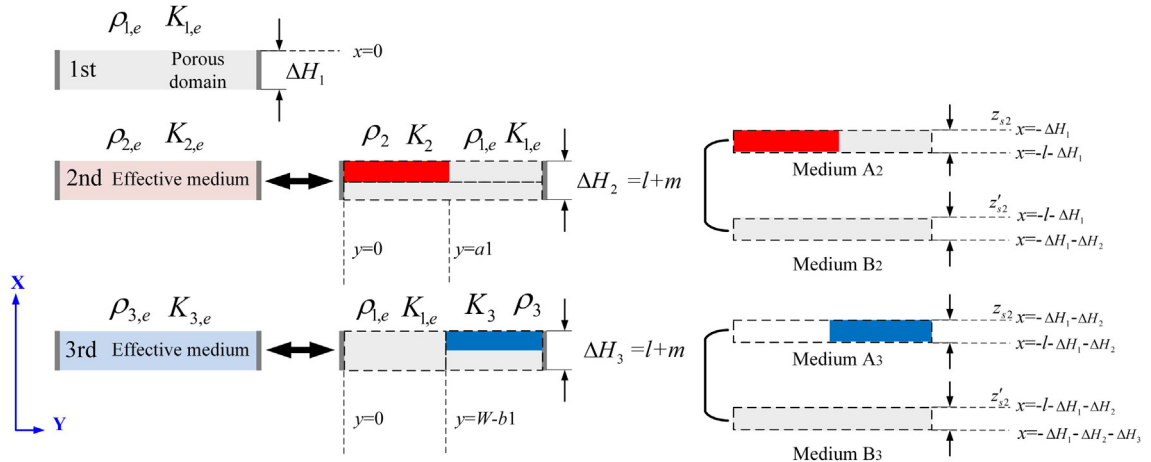


Fig. 8. Effective medium model of the composite *meta*-porous structure.

$$\rho_{2,e,x} = \rho_{1,e}/\eta_2 \quad (16)$$

$$K_{2,e} = K_{1,e}/\eta_2 \quad (17)$$

where η_i ($2 \leq i \leq 19$) is the bulk fraction of the porous sponge in medium A_i . In medium A_2 , due to the effect of the lateral board, the equivalent sound velocity $c_{2,e,x}$ and equivalent impedance $z_{2,e,x}$ of the propagating wave in the X-direction are:

$$c_{2,e,x} = \sqrt{K_{2,e}/\rho_{2,e,x}} = c_{1,e} \quad (18)$$

$$z_{2,e,x} = \rho_{2,e,x} c_{2,e,x} = z_{1,e}/\eta \quad (19)$$

hence, the equivalent surface impedance z_{s2} and z'_{s2} of porous parts in medium A_2 and B_2 are related to satisfying Eq. (19). Similarly, since the bottom of the composite *meta*-porous structure in Figs. 7 and 8 is the rigid boundary the corresponding impedance is infinite so that one can calculate z_{s3} – z_{s18} as

$$z'_{19,e} = -jz_{1,e} \cot(k_{19,e} \cdot l) \quad (20)$$

Therefore, the sound reflection coefficient $R(\omega)$ and sound absorption coefficient $\alpha(\omega)$ are as follows:

$$R = \frac{z_{s1} \cos \phi - z_0}{z_{s1} \cos \phi + z_0} \quad (21)$$

$$\alpha = 1 - |R|^2 \quad (22)$$

Fig. 9 compares the results between the TM combined with the equivalent medium model and the FEM. Under normal incidence angle, the peak and valley value of composite *meta*-porous structure with the fixed boundary at middle and low frequency correspond well. Compared with the FEM, the TM results have no significant valley values at high frequencies. For the composite *meta*-porous structure and H3 model calculated with the large unit cell, in the case of normal incidence angle, the results of TM and FEM verify each other. For the oblique incidence angle, the deviation is no more than 5%, within the allowable engineering range. This deviation may be related to the fact that the FEM considers the viscous distribution function in the medium, and the free grid also brings the deviation to the calculation results. On the whole, the proposed theoretical model is constructed by the equivalent medium model and TM, which verifies the effectiveness of the results of FEM.

4. Effect factors on the sound absorption coefficient

This session discusses the effect of four geometric dimensions $a1$, $b1$, d , and H of composite *meta*-porous structure with different boundaries on the sound absorption coefficients. When a single geometric dimension changes, other dimensions and geometric constraints remain unchanged.

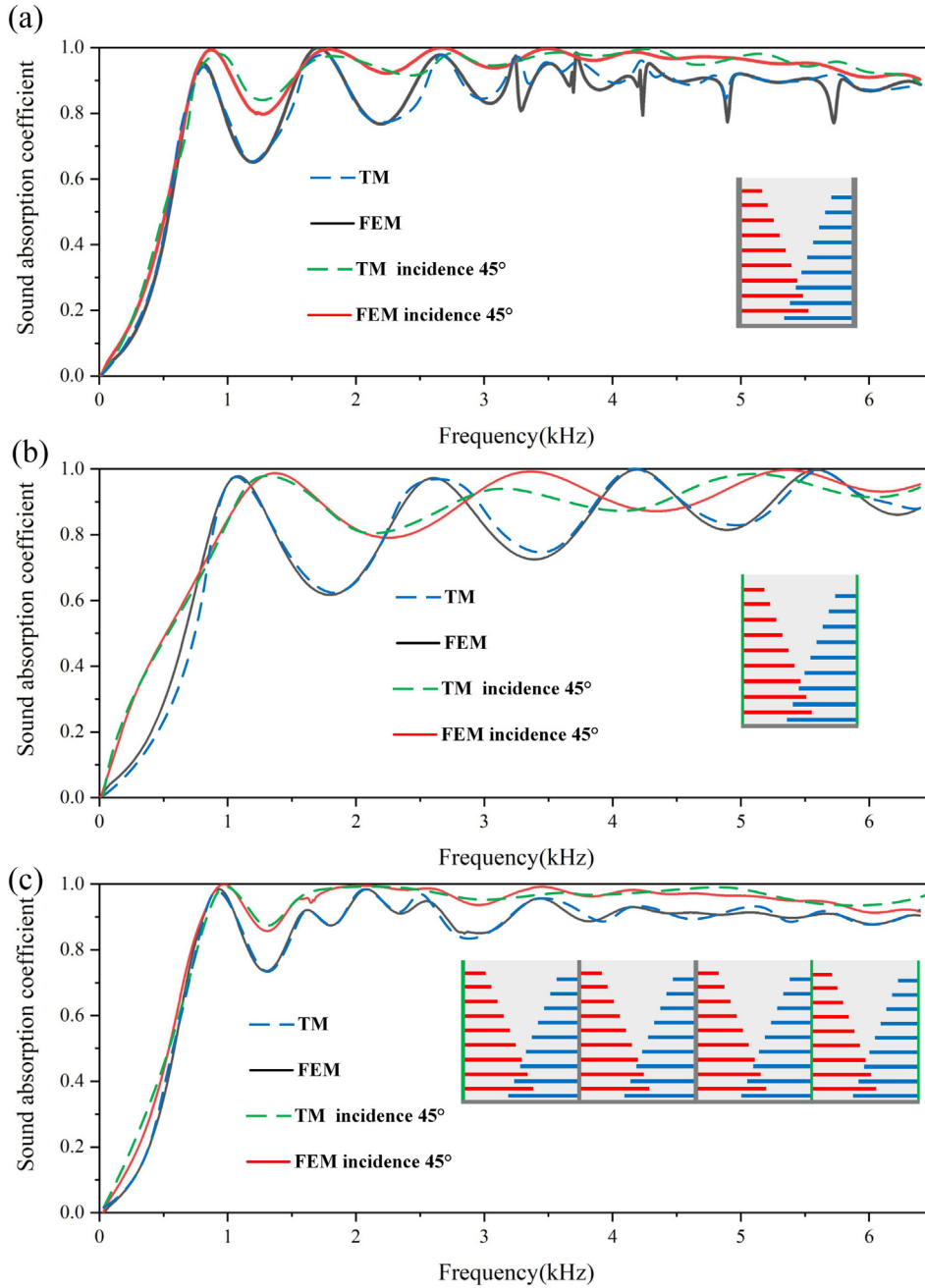


Fig. 9. Sound absorption comparison between TM (dotted line) and FEM (solid line) calculation results. Composite *meta*-porous structure (a) with fixed boundary, (b) with periodic boundary; (c) H3 model with periodic boundary.

4.1. Composite *meta*-porous structure with the fixed boundary

From Fig. 10 (a)–(d), the frequency corresponding to the maximum sound absorption coefficient is 1705 Hz, and the change of a_1 and b_1 has the similar effect on the sound absorption coefficient. When a_1 or b_1 is greater than 12 mm, sound absorption coefficients would decrease below 1 kHz and above 4 kHz. In order to maintain the high absorption coefficient, a_1 or b_1 should be less than 10 mm. Compared with H , the change of d has the more substantial effect on the sound absorption coefficient. When $d \geq 1.6$ mm, the sound absorption coefficient decreases significantly below 1 kHz; when $d \leq 0.8$ mm, the sound absorption coefficient also decreases in the ranges of 1.5–2.5 kHz and 3.4–4 kHz. The change of H weakly affects the sound absorption coefficient at the medium and low frequencies and almost does not affect high frequencies. When a_1 and

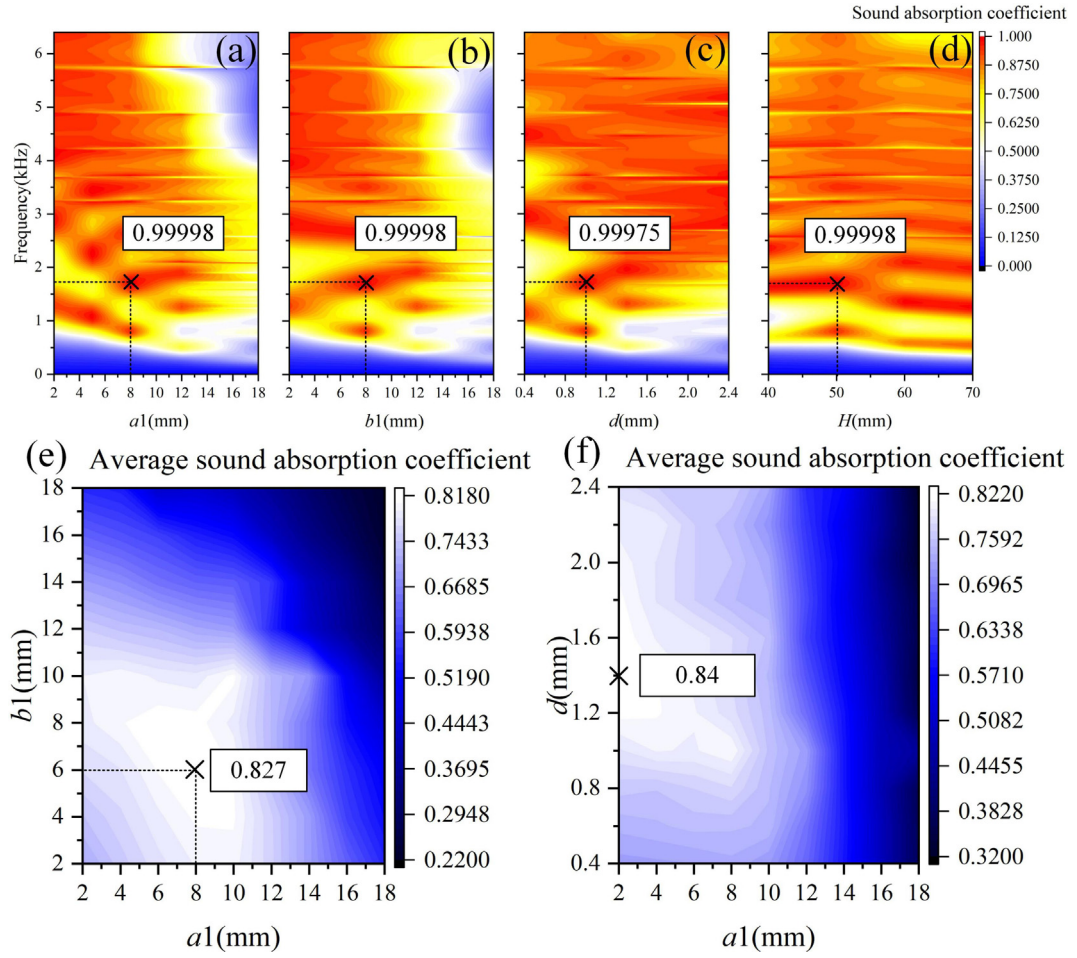


Fig. 10. Effects of $a1$ (a), $b1$ (b), d (c), and H (d) on the sound absorption coefficient of composite *meta*-porous structure with the fixed boundary. The double geometry dimension distribution of average sound absorption coefficient with $b1$ corresponding to $a1$ (a) and $a1$ corresponding to d (b) in the range of 0–6.4 kHz. The crosses indicate the coordinates of the maximum sound absorption.

$b1$ change simultaneously, the maximum value of the average absorption coefficient is 0.827, and the maximum value of the average absorption coefficient can reach 0.84. Therefore, one obtains the optimized size of the composite *meta*-porous structure with fixed boundary: $a1 = 2$ mm, $d = 1.4$ mm; Fig. 11 shows the corresponding sound absorption curve.

The average absorption coefficient is up to 0.84 in Fig. 11. When the frequency increases, the average sound absorption coefficient is also increased. In the range of 0.5–6.4 kHz, the average absorption coefficient is 0.9. From the seven distinct absorption peaks' coordinates in Table 4 and acoustic pressure distributions in Fig. 11, the sound energy absorption region with different frequencies (≥ 0.98) achieving high absorption presents spatial fluctuation characteristics in the composite *meta*-porous structure. When the frequency increases, the distribution area of the lowest acoustic pressure gradually moves from the bottom to the surface of the structure, meaning that the high absorption at different frequencies corresponds to different acoustic modes. Also, this behavior shows that the convergence effect of different lateral plates on sound energy is different at different frequencies.

4.2. Composite *meta*-porous structure with the periodic boundary

From Fig. 2, the sound absorption coefficients of the same models are different with different boundaries. Fig. 12 (a) and (b) show that the size change of $a1$ and $b1$ on the sound absorption coefficient has the same effect, with the highest sound absorption point is at 6105 Hz. When $a1$ or $b1 \geq 10$ mm, the sound absorption coefficient changes abruptly. When d increases from 0.4 to 2.4 mm, the sound absorption peak moves to the low frequency, and the highest sound absorption point is 2375 Hz. The effect of H on the sound absorption coefficient is similar to that of d , and the highest sound absorption points appear at 4815 Hz and 4625 Hz, corresponding to $H = 40$ mm and $H = 50$ mm, respectively. For the simultaneous change of $a1$ and $b1$, the maximum value of the average sound absorption coefficient is 0.79, and Fig. 12(e) shows the cor-

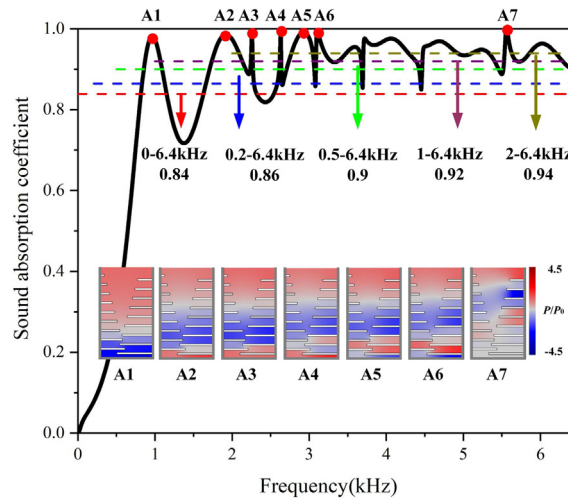


Fig. 11. Sound absorption optimization results of composite *meta*-porous structure with the fixed boundary.

Table 4

Coordinates of points in Fig. 11.

Points	A1	A2	A3	A4	A5	A6	A7
Frequency (Hz)	960	1919	2258	2632	2927	3105	5562
Sound absorption coefficient	0.98	0.99	0.99	0.99	0.99	0.99	0.99

responding sizes. When $a1$ and d change, the maximum value of the average absorption coefficient rises to 0.81. According to the results in Fig. 12(e) and (f), the optimal size of composite *meta*-porous structure with periodic structure is: $a1 = 2$ mm, $b1 = 14$ mm, $d = 2$ mm.

Fig. 13 shows the corresponding sound absorption curve and the average absorption coefficient in each different frequency range and Table 5 shows the coordinates of points B1–7. Because the first sound absorption valley widens, the overall sound absorption is not as good as that in Fig. 11, but the first sound absorption peak B1 locates at 863 Hz, 97 Hz lower than point A1. From the acoustic pressure distribution of the seven sound absorption peaks B1–B7, the spatial fluctuation characteristics with frequency change are more abundant than that of points A1–A7. Compared with the red line in Fig. 4 (a), the optimization based on the average sound absorption coefficient can only improve the high-frequency sound absorption performance and generate more absorption peaks. However, it makes the first sound absorption valley fall and wide, thus generating an unfavorable condition for the harmful noise suppression at middle and low frequencies. (See Fig. 4)

4.3. H3 model with the periodic boundary

According to the calculation results (Fig. 4 (b)), the hybrid composite *meta*-porous structure achieves better sound absorption performance. Therefore, it is necessary to discuss the effect of the geometric dimensions of the H3 model, as shown in Fig. 14. When $a1$ or $b1 = 6$ mm, the maximum sound absorption occurs at 2335 Hz, while when $a1$ or $b1$ increases, the sound absorption coefficient of the whole frequency is decreasing. When $d = 0.8$ mm, the maximum absorption point is at 2385 Hz, and the sole change of d affects the sound absorption coefficient significantly in the range of 1–2.5 kHz. Similarly, changing H moves the sound absorption peak slightly to the low frequency. When $a1 = 2$ mm and $b1 = 8$ mm, the average sound absorption coefficient can be as high as 0.83, and when $a1 = 2$ mm, and $d = 1.6$ mm, the value can be as high as 0.856. Therefore, the optimized sound absorption curve of the H3 model in Fig. 15 corresponds to the following optimized size: $a1 = 2$ mm, $b1 = 8$ mm, $d = 1.6$ mm.

Compared with Fig. 13, the first sound absorption valley in Fig. 15 significantly improves while keeping its excellent high-frequency sound absorption characteristics. However, the first and second sound absorption peaks C1 and C2 decrease to 0.94 and 0.95 (Table 6), respectively. In the 0.5–6.4 kHz, 1–6.4 kHz and 2–6.4 kHz, range, the average absorption coefficient is as high as 0.919, 0.94 and 0.955, respectively. From the acoustic pressure distribution of points C1–C7, the sound absorption peaks of the H3 model are the results of considering two kinds of boundaries. The lattice size of the large unit cell is 120 mm, and the thickness is 50 mm. Hence, this composite absorber in the periodic arrangement meets the requirements of noise and vibration reduction in large-scale space.

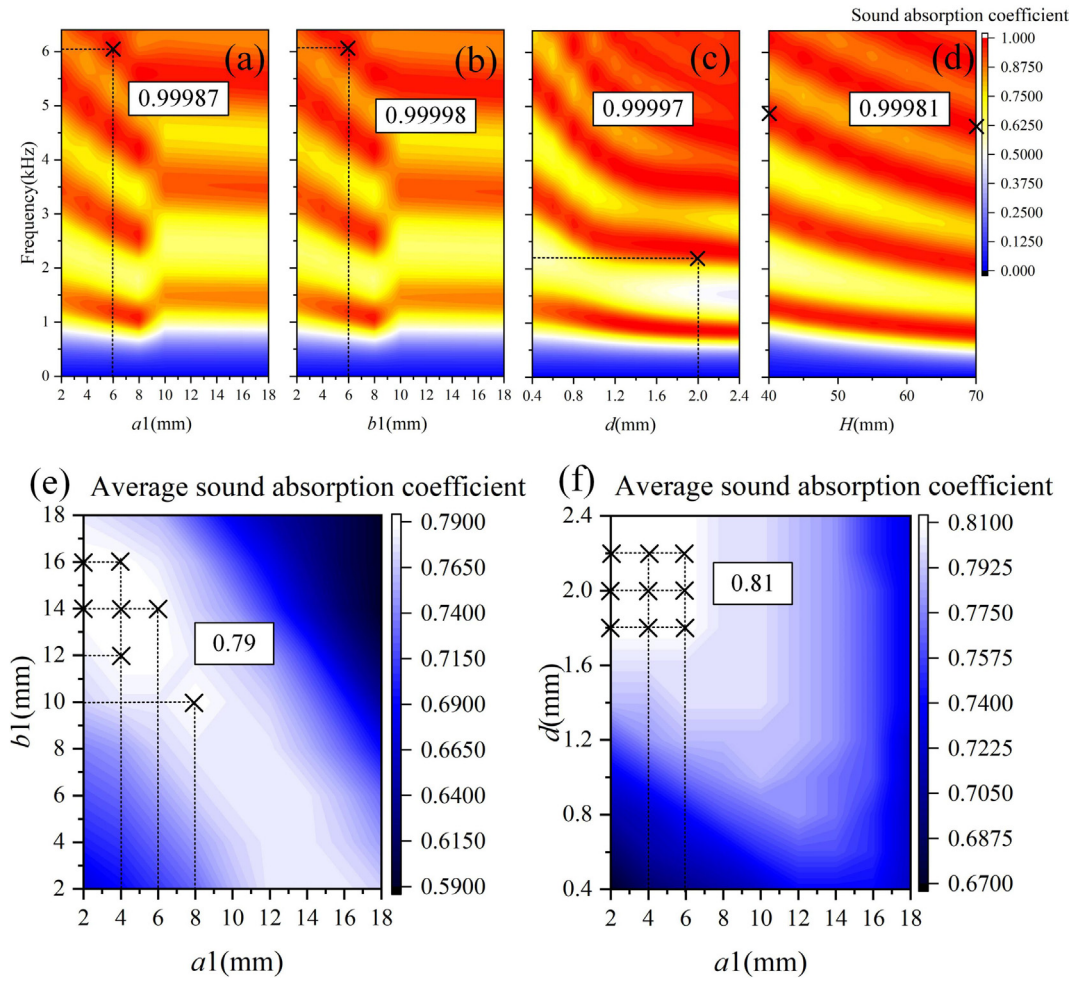


Fig. 12. Effects of a_1 (a), b_1 (b), d (c), and H (d) on the sound absorption coefficient of composite meta-porous structure with periodic boundary. The double geometry dimension distribution of average sound absorption coefficient with b_1 corresponding to a_1 (a) and a_1 corresponding to d (b) in the range of 0–6.4 kHz. The crosses indicate the coordinates of the maximum sound absorption.

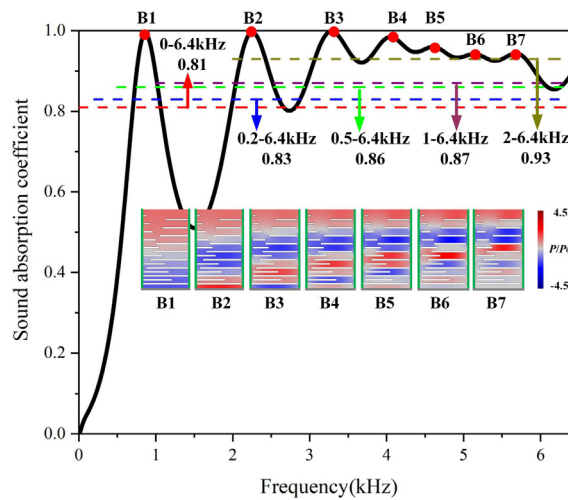


Fig. 13. The sound absorption optimization results of composite meta-porous structure with the periodic boundary.

Table 5

Coordinates of points in Fig. 13.

Points	B1	B2	B3	B4	B5	B6	B7
Frequency (Hz)	863	2255	3280	4046	4604	5134	5671
Sound absorption coefficient	0.99	0.99	0.99	0.98	0.96	0.94	0.94

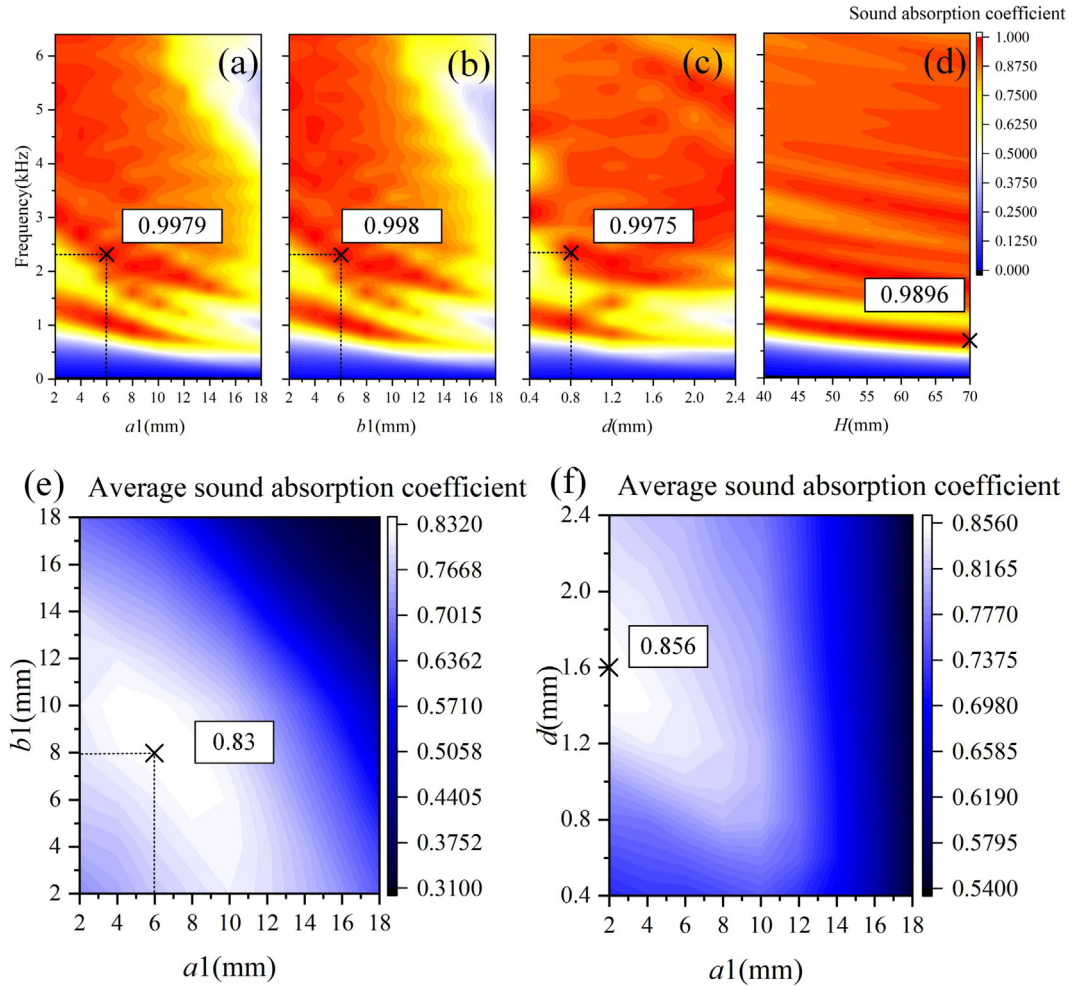


Fig. 14. Effects of $a1$ (a), $b1$ (b), d (c), and H (d) on the sound absorption coefficient of H3 model. The double geometry dimension distribution of average sound absorption coefficient with $b1$ corresponding to $a1$ (a) and $a1$ corresponding to d (b) in the range of 0–6.4 kHz. The crosses indicate the coordinates of the maximum sound absorption.

5. Test sample

5.1. Sample preparation

In order to verify the actual sound absorption performance of composite *meta*-porous structure, the small size sample for acoustic impedance tube is made, as shown in Fig. 16. Lateral plates and fixed boundaries are prepared by 3D printing system (nanoArch® P130), and their printing materials are photosensitive resins. Porous sponge is provided by Shenzhen jiexin rubber sponge product Co., LTD, the main component is high polymer polyurethane fiber.

5.2. Acoustic parameters of JCA model

The static flow resistance σ of porous sponge is tested by acoustic impedance tube (Brüel & Kjær 4206). According to the characteristic impedance z_1 and wave number k_1 of the sample, the expression of steady flow resistance [52] can be

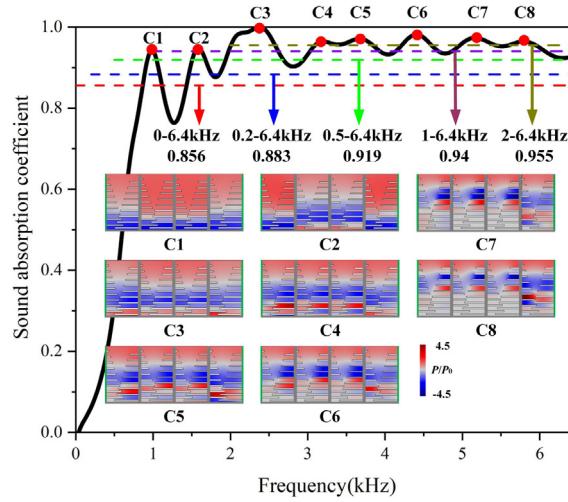


Fig. 15. The sound absorption optimization results of H3 model.

obtained: $\sigma = -\lim[\text{Im}(z_1 k_1)]$. Here we set the sample thickness for the static flow resistance to be h . When the sound absorption end of the impedance tube is the rigid wall in the test, the acoustic impedance z_L of the back cavity can be calculated by the distance L between the sample and the rigid wall:

$$z_L = -j\rho c / \tan(k_1 L) \quad (23)$$

The surface impedance z_{s1} of the sample can be calculated by the transfer function method [53], and the surface impedance z_{s1} of the sample is related to the characteristic impedance z_1 of the sample are [54]:

$$z_1 = -jz_{s1} / \tan(h \cdot k_1) \quad (24)$$

With the help of the acoustic impedance transfer formula in tube, there are the following relationships:

$$z_s = z_1 \frac{z_L + jz_1 \tan(h \cdot k_1)}{z_1 + jz_L \tan(h \cdot k_1)} \quad (25)$$

If the distance L changes from 0- h , and this relationship is rewritten as follows:

$$z'_s = z_s \frac{z_L - z_s \tan^2(h \cdot k_1)}{z_s + z_L} \quad (26)$$

The wave number k_1 can be obtained by the following formula:

$$k_1 = \pm \frac{1}{h} \sqrt{\frac{z_L}{z_s} - \frac{z'_s(z_s + z_L)}{z_s^2}} \quad (27)$$

when the thickness of test sample is far less than the wave length of air in the acoustic impedance tube, and the following approximation can be obtained:

$$z_1 k_1 = \frac{j[z_L - z'_s(z_s + z_L)/z_s]}{h} \quad (28)$$

Fig. 17 (a) and (b) show the above testing principle of static flow resistance and actual test site. Test results is 3105 Pa·S/m², and the error of flow resistance calculated by simulation is within 9.2%.

In order to get accurate porosity ϕ of porous sponge, the industrial computed tomography scanning (industrial CT) is adopted. The scanning sample size is 10 mm × 10 mm × 10 mm, phoenix v|tome|x s scanning system is provided by Shanghai Yinghua Inspection and Testing Co., LTD. In Cartesian coordinate system, 400 X-ray scans were made in three directions of 10 mm × 10 mm × 10 mm samples. Then, by calculating the porosity of each slice, the porosity ϕ of the porous sponge sample is 97.54%, and the error with the simulation data is 1.6%. From Fig. 18(a)-(c), it can be seen that the maximum pore

Table 6

Coordinates of points in Fig. 15.

Points	C1	C2	C3	C4	C5	C6	C7	C8
Frequency (Hz)	982	1589	2379	3201	3675	4398	5171	5794
Sound absorption coefficient	0.94	0.95	0.99	0.96	0.97	0.98	0.97	0.96

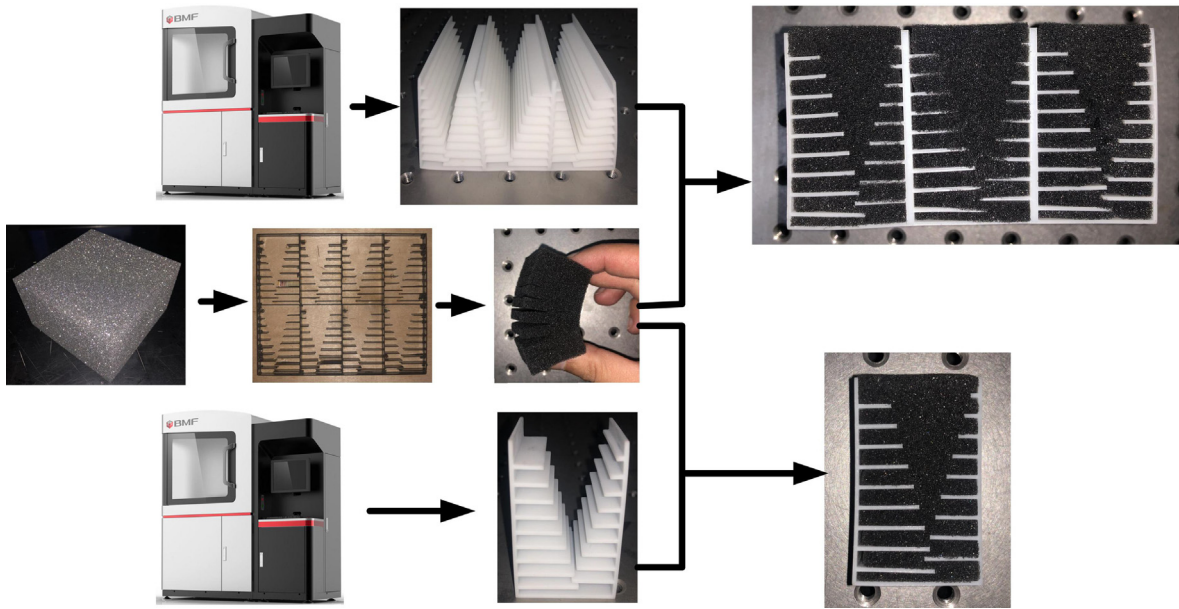


Fig. 16. The preparation process of the test sample with fixed boundary.

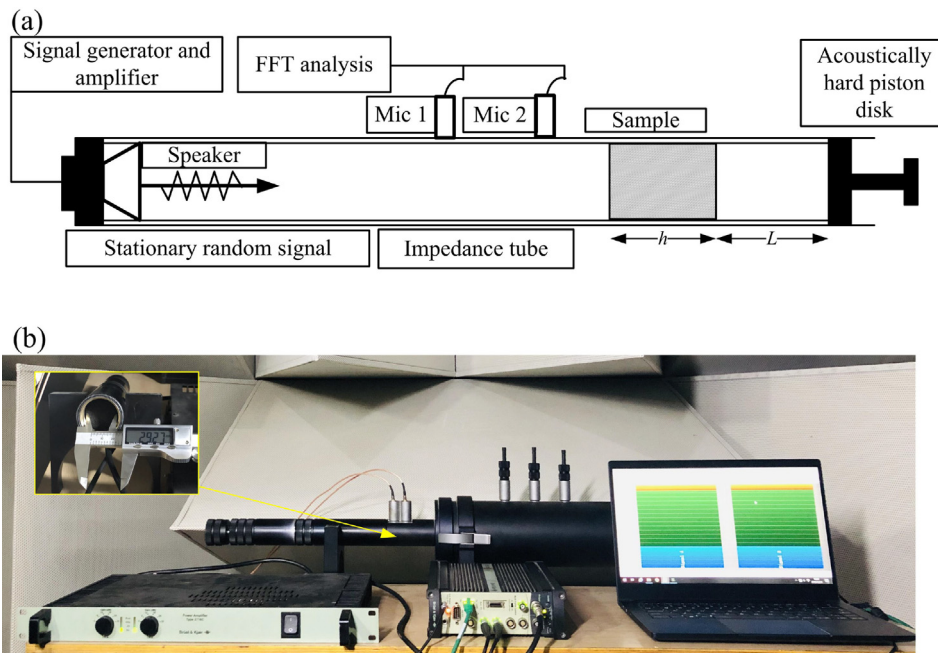


Fig. 17. Acoustic measurement method (a) and test site (b) for flow resistance test in the acoustic impedance tube.

diameter in porous sponge could exceed 1.5 mm, and the their micro morphology characteristics of pores are close to regular polygon. In the three-dimensional space, these pores are evenly distributed and penetrate each other to form the network. Air occupies most of the volume of sponge. Fig. 18(d)–(h) shows that the porous sponge used in the test belongs to the hollow structure with rigid frame.

The orientation of the pores relative to the incident sound field has an effect on the sound propagation, which is represented by the parameter tortuosity, denoted α_∞ . When the pores in porous sponge are large, tortuosity α_∞ can be deduced from the real part of the characteristics impedance [55]. Above some critical frequency, the real part of the characteristic impedance $\text{Re}(Z_c)$ approaches asymptotically to a limit value [56]:

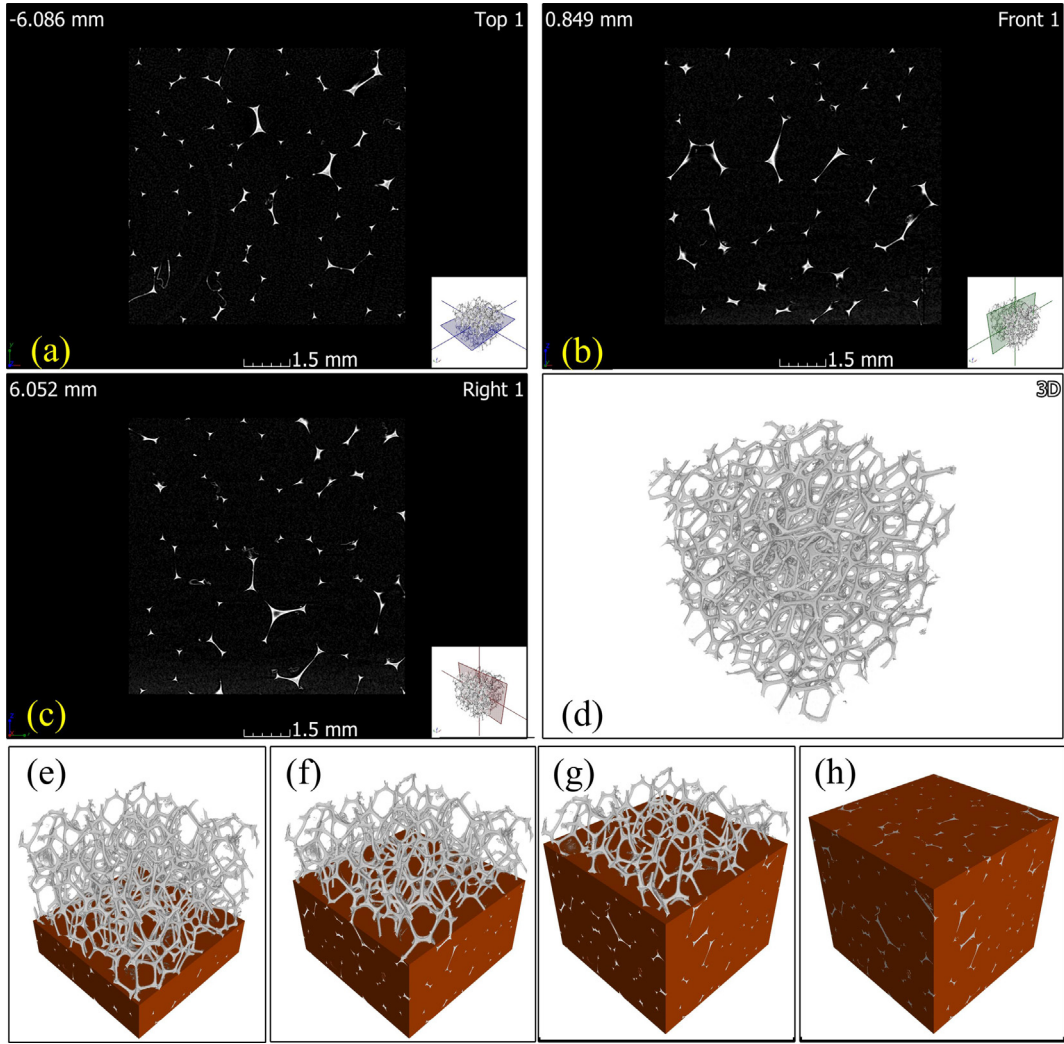


Fig. 18. The industrial CT image of porous sponge. XY, XZ, YZ directional slices (a)–(c) and 3D scanning image of porous sponge (d). (e)–(h) are the 3D rendering of the porosity analysis region. The white grid skeleton represents the sponge fiber, and the orange cube represents the air region. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$\alpha_{\infty} = \text{Re} \left(\frac{Z_c}{\rho_0 c_0} \right)^2 \phi^2 \bigg|_{\omega \rightarrow \infty} \quad (29)$$

The test equipment is still as shown in Fig. 17 (b). Tortuosity is $\alpha_{\infty} = 1.06$ after repeated calculation, which is due to the microscopic velocity.

Viscous and thermal behavior of porous sponge is affected by the shapes of pores. It can be found that viscous characteristic length Λ for simple pore shapes using the following formulation [57]:

$$\Lambda = \frac{1}{s} \sqrt{\frac{8\eta\alpha_{\infty}}{\phi\sigma}} \quad (30)$$

where s is a constant. For most porous absorbers, s lies between 0.3 and 3, taking the follow values 1, 1.07, 1.14, for circular, square, and triangular pores, respectively, and 0.78 for slits. $\eta = 1.87 \times 10^{-5}$ Pa·s is the viscosity loss of air. Thermal characteristic length $\Lambda' \cong 2\Lambda$ and $s = 1$ can be adopted to derive simpler relations for sound propagation in rigid framed fibrous materials [58]. Hence, viscous and thermal characteristic lengths Λ and Λ' are 222.24 μm and 444.48 μm , respectively. The test results of static flow resistance σ and porosity ϕ , and the other three acoustic parameters, are all the main parameters affecting the acoustic performance in JCA model, are not much different from those in FEM simulation [45,46]. Therefore, the actual porous sponge samples can be adopted to carried out the sound absorption test.

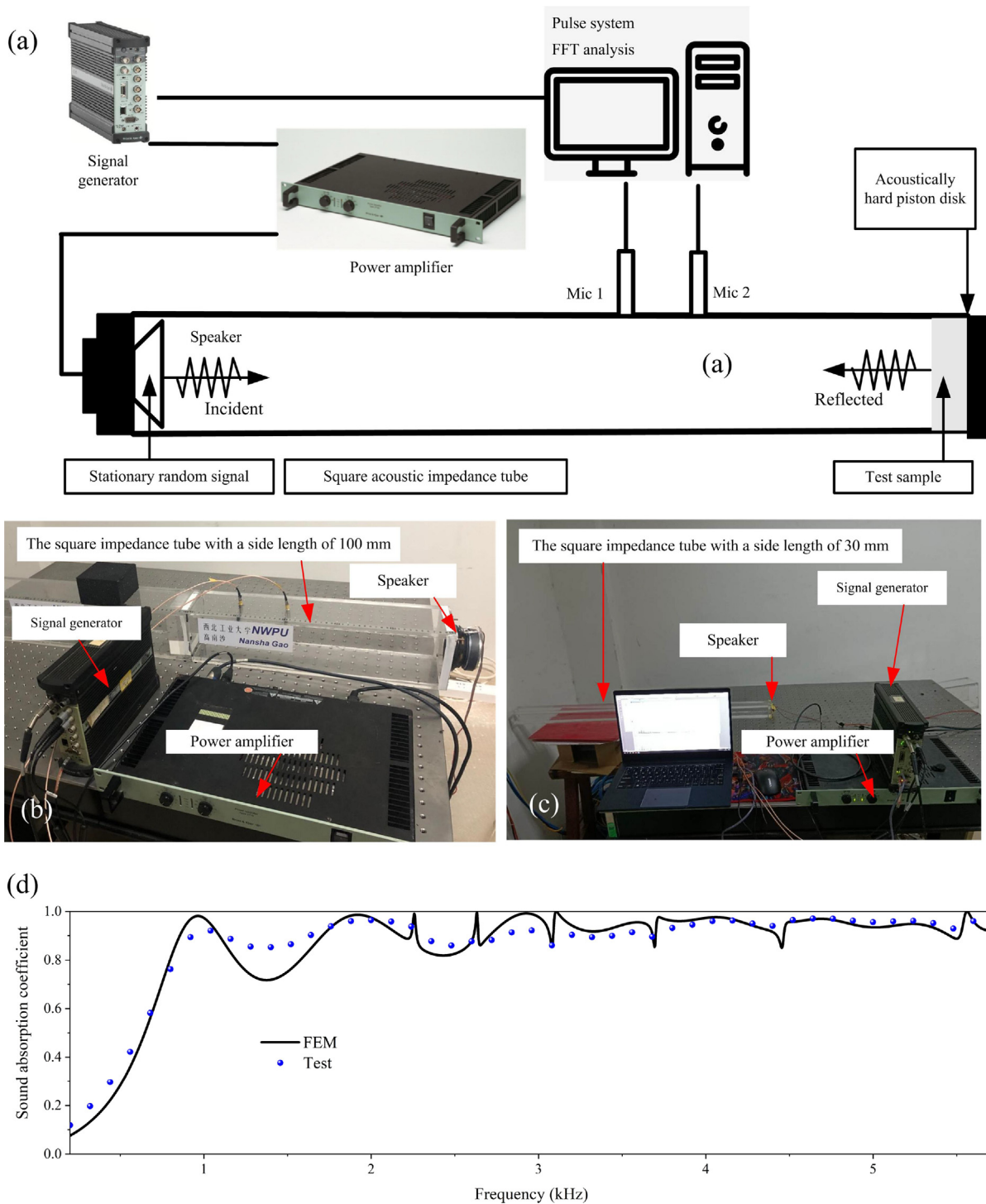


Fig. 19. (a)–(c) are the test system of sound absorption coefficient in impedance tube, (d) is the comparison between test and FEM results.

6. Sound absorption test

This test principle of sound absorption coefficient in impedance tube is carried out according to the international standard according to ISO 10534-2 [53]. Combined transfer function H_{12} of the total sound field of two microphones and the distance between two microphones, normal absorption coefficient are calculated by transfer function method. We build the two

square tube with different sizes. In order to ensure that there is no high-order wave in the square tube, the maximum effective frequency is 1.715 kHz for the side length 100 mm and 5.716 kHz for the side length 30 mm [54]. The testing principle and corresponding process are shown in the Fig. 19 (a)–(c). Limited by the physical limits of speakers, minimum effective frequency is 0.2 kHz. In Fig. 19 (d), the test data differ little from the results given by the FEM and TM, and the first sound absorption peak has a high coincidence, which proves the rationality of the design in this study and the correctness of the theoretical solution and the FEM calculation.

7. Conclusions

This study proposes composite *meta*-porous structure, investigates its sound absorption characteristics in detail. Regardless of fixed or periodic boundary, the theoretical models adopt the TM method combined with the effective medium model to calculate the sound absorption coefficient, and its validity is proved by FEM results. Relative to pure porous sponge and contrast models S1(S2), the lateral plate in composite *meta*-porous structure could promote the whole sound absorption. Normalized surface impedance, and reflection coefficient on the complex plane further illustrates the phase change of the reflection coefficient is the critical condition to achieve a high sound absorption. Complex plane analysis of the reflection coefficient further shows that the adjustment of the structural design comprehensively from the balance between the loss, the leakage, and the isolated resonance coupling strength, to achieve the better absorption target. This study also discussed the change of four geometry parameters under three different boundary conditions in detail. The lengths of left (right) lateral plate a_1 (b_1) and length difference d between upper and lower lateral plates are the most important factor. Results of self-built impedance tube test system demonstrates the correctness of FEM in range of 0.2–5.716 kHz. The conclusions has potential application in the composite structure design of porous materials, and is suitable for the sound absorption design in confined space. Our future study would expand to the design of broadband sound absorber in free sound field.

CRediT authorship contribution statement

Nansha Gao: Investigation, Conceptualization, Funding acquisition, Supervision. **Jianguo Wu:** Visualization, Software. **Kuan Lu:** Methodology. **Haibin Zhong:** .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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